C S Vaishakh

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Professional Summary

AI & Data Science undergraduate student with a strong interest in backend development, systems architecture, and machine learning. Experienced in building RESTful APIs with Go (Fiber) and Node.js (Express), managing relational databases like PostgreSQL, and developing full-stack web applications.

Education

St. Joseph's College of Engineering and Technology, Palai Kottayam, Kerala
B.Tech in Artificial Intelligence and Data Science — APJ Abdul Kalam Technological University *2023 – 2027*

Projects

Quickhand December 2025 – Present

- Implemented role-based access control (RBAC) across two user roles, securing Fiber API routes and verifying sessions with stateless JWTs.
- Engineered a high-performance monorepo using a Go backend (Fiber v3) and a React frontend (TanStack Start) orchestrated via Bun.
- **Technologies**: Go, Fiber v3, React, TanStack Start, TypeScript, Tailwind CSS, PostgreSQL, GORM, Docker

Snip July 2026

- Built a full-stack URL shortener with a Go (Gin) REST API backed by PostgreSQL and Redis, and a React/TypeScript dashboard for managing short links.
- Implemented JWT-based authentication and collision-safe short code generation, with a Redis-backed cache in front of PostgreSQL on the public redirect path.
- Deployed the React frontend on Vercel and the Go backend on an AWS EC2 instance behind Caddy, with GitHub Actions CI building and vetting both apps on every push.
- **Technologies**: Go, Gin, GORM, PostgreSQL, Redis, JWT, React, TypeScript, Vite, Tailwind CSS, Docker, AWS EC2, Vercel

ONT Monitor April 2026

- Built a Go-based microservice using Fiber v3 and GORM to ingest and process real-time status change events for ONTs.
- Implemented thread-safe in-memory queues and a background worker pool to evaluate outages and prevent race conditions.
- Integrated Twilio API to dispatch automated outage and recovery WhatsApp alerts to registered customers upon state transitions.
- **Technologies**: Go, Fiber v3, GORM, PostgreSQL, Twilio API, Docker

PortfolioIQ December 2025 – April 2026

- Engineered a privacy-preserving portfolio rebalancing platform structured as a monorepo using npm workspaces.
- Designed a client-side training pipeline using TensorFlow.js to train local logistic regression models on a 12-dimensional feature vector, eliminating raw data uploads.
- Implemented the Federated Averaging (FedAvg) algorithm on the Express.js aggregator to synchronize local updates with a FastAPI scikit-learn model.
- **Technologies**: Next.js, Express.js, FastAPI, TensorFlow.js, Drizzle ORM, PostgreSQL, JWT, bcrypt, Python, TypeScript, Turborepo, Docker

Technical Skills

Programming: Python, TypeScript, Go, Java, JavaScript
Web & Backend: Next.js, React, Express.js, Fiber v3, Gin, FastAPI, Node.js, TanStack Start, REST APIs
Databases & ORMs: PostgreSQL, MongoDB, Redis, GORM, Drizzle ORM, Mongoose
AI / ML: TensorFlow.js, Feature Engineering, Model Evaluation, scikit-learn, NumPy, Pandas
Tools: Git, GitHub Actions, Docker, AWS, Vercel, Turborepo, Bun, Postman, Linux, VS Code